Software Requirements Specification

for

<Blood Bank Management System>

Version 1.0 approved

Prepared by<Riya Patil>

<KIT's College of Engineering>

<29 th July ,2026>

Table of Contents

Table of Contents ii
Revision History ii
1. Introduction 1
1.1 Purpose 1
1.2 Document Conventions 1
1.3 Intended Audience and Reading Suggestions 1
1.4 Product Scope 1
1.5 References 1
2. Overall Description 2
2.1 Product Perspective 2
2.2 Product Functions 2
2.3 User Classes and Characteristics 2
2.4 Operating Environment 2
2.5 Design and Implementation Constraints 2
2.6 User Documentation 2
2.7 Assumptions and Dependencies 3
3. External Interface Requirements 3
3.1 User Interfaces 3
3.2 Hardware Interfaces 3
3.3 Software Interfaces 3
3.4 Communications Interfaces 3
4. System Features 4
4.1 System Feature 1 4
4.2 System Feature 2 (and so on) 4
5. Other Nonfunctional Requirements 4
5.1 Performance Requirements 4
5.2 Safety Requirements 5
5.3 Security Requirements 5
5.4 Software Quality Attributes 5
5.5 Business Rules 5
6. Other Requirements 5
Appendix A: Glossary 5
Appendix B: Analysis Models 5
Appendix C: To Be Determined List 6

Revision History

Name	Date	Reason For Changes	Version

1. Introduction

1.1 Purpose

The purpose of a blood bank management system is to digitize and automate operations, manage blood inventory, and track donor information to ensure a safe and fast supply of blood. It removes paperwork and lowers human mistakes during life-saving emergencies.

1.2 Document Convention

The SRS (Software Requirement Specification) documentation follows a standardized format to ensure clarity and consistency. It uses structured sections such as introduction, problem statement, problem description and modules. Clear headings, simple formal language, and proper numbering are maintained throughout. Standard naming conventions, well-defined terms, and consistent formatting are used to make the document easy to understand for all stakeholders.

1.3 Intended Audience and Reading Suggestions

The SRS document for the Blood Bank Management System is intended for developers, project managers, testers, administrators, hospital staff, donors, patients& blood bank staff. It provides clear system requirements and features. Readers should start with the introduction for overview and then refer to specific sections based on their needs for better understanding.

1.4 Scope

The Blood Bank Management System is designed to efficiently manage and maintain records of blood donors, recipients, and blood inventory. It enables functionalities such as donor registration, blood group management, stock tracking, and request handling. The system aims to reduce manual work, improve accuracy, and ensure quick availability of blood during emergencies. It also provides secure data storage and easy access for authorized users, making the overall blood bank operations more reliable and efficient.

1.5 References

- IEEE SRS Standards
- WHO Guidelines on Blood Bank Systems
- Relevant online resources

2. Overall Description

2.1 Perspective

The Blood Bank Management System is an independent application designed to manage blood bank operations efficiently. It interacts with users such as administrators, staff, and donors through a user-friendly interface. The system maintains a centralized database to store donor details, blood stock, and requests. It may also be integrated with external systems like hospital databases for better coordination. Overall, it works as a standalone system that improves accuracy, reduces manual effort, and ensures timely availability of blood.

2.2 Functions

- Donor registration and management
- Blood group and stock management
- Blood request and approval handling
- Search for available blood units
- Record of donors and recipients
- Admin login and user management
- Report generation and data tracking
- Notification about blood availability or shortage

2.3 User Classes and Characteristics

The Blood Bank Management System has different user classes including Administrator and Staff as primary users who manage operations and use the system frequently, and Donors, Recipients, and Hospitals as secondary users who access limited features occasionally. The Administrator and Staff are the most important users as the system mainly depends on their activities.

2.4 Operating Environment

The Blood Bank Management System will operate in a client-server environment using standard computer systems with basic hardware such as desktops or laptops. It is compatible

with operating systems like Windows or Linux and runs through a web browser or dedicated application. The system uses a database (such as MySQL) for data storage and may integrate with hospital systems or other healthcare applications to ensure smooth and efficient operation.

2.5 Design and Implementation Constraints

The Blood Bank Management System must follow data privacy and healthcare regulations, use standard hardware and technologies like Java/Python and MySQL, and ensure secure access through authentication. It should be compatible with existing hospital systems, follow coding standards, and be easy to use for staff while operating efficiently within available resources.

2.6 User Documentation

The Blood Bank Management System will include user documentation such as a user manual, online help support, and basic tutorials to guide users in operating the system. These documents will be provided in digital formats like PDF and web-based help pages, following standard documentation practices to ensure easy understanding and accessibility for all users.

2.7 Assumptions and Dependencies

The Blood Bank Management System assumes the availability of stable internet connectivity, standard hardware systems, and a secure database environment such as MySQL. It also assumes that users have basic computer knowledge and that external systems like hospital databases are accessible for integration. The project depends on third-party software tools, development platforms, and proper functioning of network and server infrastructure. Any changes or failure in these assumptions or dependencies may affect the system' s performance and requirements.

3. External Interface Requirements

3.1 User Interfaces

The Blood Bank Management System will provide a simple and user-friendly graphical interface for donors, staff, and administrators. All screens will follow a consistent design with standard colors, fonts, and navigation menus for easy use. Each interface will include input fields, action buttons such as Add, Update, Delete, and Search, along with a help option and

proper navigation like Dashboard, Donor, Blood Stock, Requests, and Reports. The system will ensure proper input validation and display clear error and success messages. Basic keyboard shortcuts may be provided for common operations. The main UI components include donor management, blood stock management, request handling, admin dashboard, and report generation. Detailed UI design will be documented separately in the user interface specification document.

3.2 Hardware Interfaces

The Blood Bank Management System (BBMS) will interact with standard hardware components such as computers, barcode scanners, printers, and database servers. The system will support devices like desktops and laptops with basic input/output hardware including keyboard, mouse, and display. Barcode scanners may be used to quickly identify blood bags and donor records, while printers will be used to generate reports and receipts. The software will communicate with hardware through standard operating system drivers and interfaces, ensuring smooth data input and output operations. Data exchange between the system and hardware will be simple and secure, using standard communication protocols supported by the operating system. The system is designed to work efficiently on commonly available hardware without requiring specialized equipment.

3.3 Software Interfaces

The Blood Bank Management System will interface with various software components including the operating system (such as Windows or Linux), database management system (such as MySQL), and development tools or libraries used for building the application. The system will store and retrieve data like donor details, blood stock, and request records from the database. Input data includes user entries such as donor registration and blood requests, while output data includes reports, availability status, and notifications. The system will use standard APIs and database connectivity (e.g., JDBC) for communication between application and database. Data sharing across modules like donor management, stock management, and reporting will be done through a centralized database to ensure consistency. All communications will follow secure and reliable protocols supported by the system, and any specific implementation constraints will follow standard software development practices.

3.4 Communications Interfaces

The Blood Bank Management System will support communication through web browsers and network servers using standard protocols such as HTTP/HTTPS. The system may also use e-mail services to send notifications regarding blood availability, donor registration, and request status. Data will be transmitted in structured formats such as forms and database records. Secure communication will be

ensured using encryption techniques (HTTPS) to protect sensitive data like donor details. The system will operate on standard network speeds and ensure proper synchronization between client and server to maintain accurate and updated information. All communication will follow common web standards to ensure reliability and security.

4. System Features

The Blood Bank Management System (BBMS) includes key features such as donor registration, blood stock management, blood request handling, and report generation. It also provides an admin dashboard to manage users and monitor system activities. These features help in efficient data management and smooth operation of the system.

4.1 System Feature 1

The Blood Bank Management System (BBMS) includes features such as Donor Management for storing donor details, Blood Stock Management to track availability of different blood groups, Blood Request Processing to handle requests from hospitals and patients, and Report Generation to provide necessary records and summaries. It also includes an Admin Dashboard for monitoring and controlling system activities, ensuring efficient and smooth operation of the system.

4.1.1 Description and Priority

The Blood Bank Management System (BBMS) includes features like Donor Management, Blood Stock Management, Blood Request Processing, Admin Dashboard, and Report Generation. Donor, Stock, and Request features are high priority as they are essential for system operation, while Admin Dashboard is medium priority and Reports are priority. High-priority features provide maximum benefit and have higher impact if not implemented.

4.1.2 Stimulus/Response Sequences

In the Blood Bank Management System (BBMS), the sequence of user actions and system responses defines how the system behaves. For example, in Donor Registration, the user enters donor details and submits the form, after which the system validates the data and stores it in the database, displaying a success message. In Blood Search, the user selects a blood group, and the system displays available units. In Blood Request Processing, the user submits a request, and the system checks availability, approves or rejects the request, and updates the stock. In Admin operations, the admin logs in, manages records, and the system updates and confirms changes. These interactions ensure smooth communication between the user and the system.

4.1.3 Functional Requirements

The Blood Bank Management System must support essential operations such as managing donor data, tracking blood stock, processing requests, and generating reports. The system should ensure accurate data entry, quick retrieval, and proper handling of errors or invalid inputs to maintain reliability and efficiency.

REQ-1: The system shall allow users to register and store donor details.
REQ-2: The system shall validate all input data and display error messages for invalid or missing fields.
REQ-3: The system shall maintain and update blood stock information in real time.
REQ-4: The system shall allow users to search blood availability by blood group.
REQ-5: The system shall process blood requests and update stock accordingly.
REQ-6: The system shall generate reports for donors, stock, and transactions.
REQ-7: The system shall provide admin access for managing users and records.
REQ-8: The system shall display appropriate success or error messages for each operation.

5. Other Nonfunctional Requirements

5.1 Performance Requirements

The Blood Bank Management System should provide quick response to user actions within a few seconds. It must handle multiple users efficiently and update blood stock data in real time. Fast and reliable performance is required to ensure smooth operation and timely access to information.

5.2 Safety Requirements

The Blood Bank Management System (BBMS) should ensure data security and prevent unauthorized access using authentication and access control. It must validate inputs to avoid errors, provide data backup to prevent loss, and follow healthcare data protection standards to ensure safe and reliable operation.

5.3 Security Requirements

The Blood Bank Management System (BBMS) should ensure data security using user authentication and role-based access control. It must protect sensitive data through

encryption and follow healthcare data privacy standards to maintain confidentiality and integrity.

5.4 Software Quality Attributes

The Blood Bank Management System (BBMS) should be reliable, user-friendly, and easy to maintain. The system should provide accurate data processing, quick response time, and high availability for users. It should be flexible enough to support future updates, portable across common platforms, and easy to test and modify. The system should ensure usability so that users can operate it with minimal training.

5.5 Business Rules

The Blood Bank Management System (BBMS) should follow defined operating rules where different users have specific roles and permissions. Donors can register and update their details, staff can manage blood stock and process requests, and administrators can manage users and system settings. Only authorized users should access sensitive information and perform critical operations.

6. Other Requirements

The Blood Bank Management System (BBMS) should maintain a secure and organized database for storing donor, patient, and blood stock information. The system should support future enhancements, follow healthcare data management rules, and ensure data privacy. It should be designed for easy maintenance, scalability, and possible reuse of modules in similar healthcare applications.

Appendix A: Glossary

BBMS stands for Blood Bank Management System, which manages donor records, blood stock, and requests. User refers to people who interact with the system, while Admin is responsible for managing system operations. Database is the organized storage of all system-related information.

Appendix B: Analysis Models

The BBMS uses models such as Data Flow Diagram (DFD), Entity Relationship Diagram (ERD), Class Diagram, and State Transition Diagram to represent data flow, database relationships, and system behavior. These models help in designing and developing the system efficiently.

Appendix C: To Be Determined List

The following details are yet to be finalized and marked as TBD. These include specific database configurations, security certifications, external system integrations, and future enhancement requirements. These references will be reviewed and updated d*uring the development phase.*